EMERSON AREHART

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EDUCATION

Ph.D., Biology **2021**
University of Utah, Salt Lake City, UT
Dissertation: Quantifying the Fitness Benefit of Learning
Advisor: Frederick Adler

B.A., Mathematics (minor in Journalism) **2008**
University of Nevada, Reno, NV

APPOINTMENTS

Conservation Social Scientist **2025–Present**
The Peregrine Fund, Boise, ID (remote)

- Inaugural social scientist at the organization; lead social science initiatives across all programs worldwide.
- Design qualitative and quantitative research, including household and attitude surveys, livelihoods assessment, and program monitoring and evaluation.
- Current research includes human–wildlife conflict and trafficking of wildlife parts in East Africa and the Caribbean, lead-free ammunition adoption among U.S. hunters, methods for reliable survey design, and evidence synthesis on the effectiveness of conservation interventions.
- Supervise in-country research teams and mentor an international group of early-career researchers.
- Apply research insights and social science principles to conservation program design.

Faculty by Courtesy **2025–Present**
College of Earth, Ocean, and Atmospheric Sciences, Oregon State University, Corvallis, OR

Postdoctoral Fellow (joint appointment), Levin Lab / High Meadows Environmental Institute **2024–2025**
Princeton University, Princeton, NJ

Postdoctoral Fellow, Social and Cultural Evolution Working Group **2022–2025**
University of Pennsylvania, Philadelphia, PA

PUBLICATIONS

Peer-reviewed

1. **Arehart, E.**; Gilbert, S.Z.; Rasolofoniaina, B.J.R.; Anjaranirina, E.J.G.; Ravaoliny, L.; Randriamady, H.J.; Golden, C.D. (2026). Seasonality of fruiting phenology, hunting behaviour and taste preferences in Madagascar's Makira Protected Area. *People and Nature*, 8, 935–944. doi:10.1002/pan3.70269. Cover article.
2. **Arehart, E.**; Adler, F.R. (2023). A minimal model of learning: quantifying the cost and benefit of learning in changing environments. *Proceedings of the Royal Society B*, 290(2005), 20231084. doi:10.1098/rspb.2023.1084
3. **Arehart, E.**; Reimer, J.R.; Adler, F.R. (2023). Strategy maps: Generalised giving-up densities for optimal foraging. *Ecology Letters*, 26(3), 398–410. doi:10.1111/ele.14160
4. Meredith, H.R.; **Arehart, E.***; Grantz, K.*; et al. (2021). A coordinated strategy for a modeling-based

decision support tool for COVID-19 in Utah, USA. *Emerging Infectious Diseases*, 27(5). doi:10.3201/eid2705.203075. *Equal contribution.

5. **Arehart, E.**; Jin, T.; Daniels, B.C. (2018). Locating decision-making circuits in a heterogeneous neural network. *Frontiers in Applied Mathematics and Statistics*, 4, 11. doi:10.3389/fams.2018.00011

Preprints

- **Arehart, E.**; Akçay, E. (2025). Biased transmission drives an accuracy–consensus tradeoff in collective learning. *SocArXiv*. doi:10.31235/osf.io/wdcps_v1

GRANTS, FELLOWSHIPS AND AWARDS

2024 Data Science for Social Good Grant, PI – *Forecasting Regime Shifts in Social-Ecological Systems: The Amazon Basin as a Case Study* ($28,700)

2022 Social and Cultural Evolution Working Group Postdoctoral Fellowship, University of Pennsylvania ($170,000 over 3 years)

2020 Mountaineer Fellowship for Graduate Research, University of Utah ($14,000)

2019 Alfred J. Lotka Prize for Best Theoretical Ecology Poster, Ecological Society of America Annual Meeting

2018 Global Change and Sustainability Center Conservation Research Grant, PI – *Conservation and Sustainability of Chile's Prized Wild Edible Mushroom, Boletus loyo* ($3,000)

2016 Global Change and Sustainability Center Fellowship, University of Utah ($21,000)

INVITED PRESENTATIONS

Arehart, E. Biased Transmission Drives an Accuracy–Consensus Tradeoff in Collective Learning.
Society for Mathematical Biology Annual Meeting, Edmonton, Canada **July 2025**

Arehart, E. Quantifying the Costs and Benefits of Learning.
Keynote, Collective Behavior Meeting, Meiji University, Tokyo, Japan **January 2025**

Arehart, E. A Minimal Model of Learning: Quantifying the Cost and Benefit of Learning in Changing Environments.
Naomi Leonard Lab, Princeton University **March 2023**

Arehart, E. Phenology and Taste Preferences in Makira, Madagascar.
Golden Lab, Harvard T.H. Chan School of Public Health **December 2022**

Arehart, E. When Does the Ability to Learn Benefit Foragers?
Special Session on Mathematical Modeling of Biological and Social Systems, American Mathematical Society Western Sectional Meeting **October 2022**

Arehart, E. When Does the Ability to Learn Benefit Foragers?
Penn Symposium on Tools, Beliefs, and Behaviors, University of Pennsylvania **May 2021**

Arehart, E. Strategy Maps: How Optimal Foragers Balance Risk, Reward, Exploration, and Learning.
MindCORE Seminar Series, University of Pennsylvania **December 2020**

CONTRIBUTED PRESENTATIONS

Arehart, E. Combining Machine Learning and Process-Based Models in Ecological Prediction.
Talk. ASLO Aquatic Sciences Meeting, Palma, Mallorca **June 2023**

Arehart, E. Quantifying the Fitness Benefit of Sensory Learning as a Function of Environmental Complexity. Winner of the Alfred J. Lotka Prize.
Poster. Ecological Society of America Annual Meeting, Louisville, KY **August 2019**

Arehart, E.; Dentinger, B. Genome Sequencing and Population Genetics of an Endangered Chilean Mushroom, Boletus loyo.
Poster. Environment & Sustainability Research Symposium, University of Utah **February 2019**

Kuzma, H.A.; **Arehart, E.** Listening to Seismic Events: Is the Ear Easier to Train than the Eye?
Poster. Comprehensive Nuclear-Test-Ban Treaty Organization Science and Technology Conference, Vienna, Austria **June 2011**

STUDENTS MENTORED

- **Faith Achieng** (MSc student, Environment and Natural Resource Management, Africa Nazarene University, Kenya, 2026–Present). Community perceptions of retaliatory wildlife poisoning, vulture risk, and the role of predator-proof bomas in Laikipia, Kenya.
- **Cheyenne Jarman** (PhD student, Oregon State University, 2023–2026). Scientific machine learning: universal differential equations to forecast regime shifts in coral reef ecosystems.
- **Joseph Katz** (undergraduate, University of Pennsylvania, 2024–2025). Forecasting regime shifts in social-ecological systems: the Amazon Basin.
- **Jake Baldauf** (undergraduate, Brigham Young University, 2020–2021). COVID-19 genomic epidemiology.

ACADEMIC SERVICE AND OUTREACH

2022–Present	Reviewer: American Naturalist, Journal of Theoretical Biology, Theoretical Population Biology, Cognition, The Journal of Supercomputing
2019–2020	Chair, Graduate Student Advisory Committee, School of Biological Sciences, University of Utah
2019	Graduate Student Representative, School of Biological Sciences, at the SACNAS (Society for the Advancement of Chicanos/Hispanics and Native Americans in Science) National Conference, Honolulu, HI
2018–2019	Graduate Student Invited Speaker Organizer and Graduate Admissions Committee member, School of Biological Sciences, University of Utah
2017	STEM Ambassador, University of Utah; 8th-grade science fair mentor, Salt Lake Center for Science Education; Lead Instructor, Empowering Youth Through STEM Outreach, Glendale Public Library, Salt Lake City

SKILLS

Quantitative: mathematical modeling (dynamical systems, dynamic programming and optimal control, agent-based and network models); Bayesian and frequentist statistics; survey design and psychometrics; experimental and quasi-experimental program evaluation; machine learning; remote sensing

Software: R, Python, Git

Working across fields: connecting methods from distant areas (foraging theory to collective learning; machine learning to mechanistic models; survey methodology to conservation program design); collaborating with biologists, social scientists, engineers, and field staff from widely different backgrounds

Leadership: designing and leading international field teams; supervising graduate and undergraduate researchers; journalism-trained writer for technical and general audiences

Languages: English (native), French (basic)

SELECTED FIELD EXPERIENCE

2026 **Ridgway's Hawk program, Dominican Republic**

The Peregrine Fund. Designed and conducted household surveys and community workshops; trained enumerators; led community meetings and site visits with program staff.

2019 **Urban microbial survey, Stirling, Scotland**
Funded by Diageo. Surveyed microbial growth near industrial facilities in an urban/suburban area.

2019 **Loyo mushroom collecting expedition, Valdivia, Chile**
Funded by a Global Change and Sustainability Center grant to E. Arehart, with support from the Natural History Museum of Utah. Collected specimens of a rare mushroom used for the first genome sequencing of the species; data contributed to its IUCN Red List assessment.

2018 **Collecting expedition, Dja Faunal Reserve, Cameroon**
Funded by the University of Utah and Natural History Museum of Utah. Collected specimens in remote Congo Basin rainforest.

2017 **Beekeeping documentary, Camagüey, Cuba**
National Geographic Young Explorer Grant to Adrian Curbelo Diaz. Consulted on beekeeping techniques and honeybee biology for a documentary film crew.

2006–2007 **Wetland restoration, Nevada**
Otis Bay Ecological Consultants. Led amphibian and plant surveys on public and private restoration sites, including Ash Meadows National Wildlife Refuge, Soldier Meadows, and McCarran Ranch.

TEACHING

2017–2020 Teaching Assistant, University of Utah: Statistics for Biologists (graduate), Urban Ecology (graduate), Entomology, Mycology, Mathematical Modeling for Biologists

PRIOR RESEARCH AND PROFESSIONAL EXPERIENCE

Graduate Research Assistant, University of Utah **2017–2021**

- *Enhancing Transmission Models of COVID-19 with New Data Sources* (L. Keegan, PI, 2020). Member of Utah's COVID-19 modeling task force; population-genomic estimates of cryptic transmission.
- *Salivary Protein Influence on Taste and Feeding*, NIH R01 (A.-M. Torregrossa, PI, SUNY Buffalo, 2018–2021). Experimental design, power analysis, statistical analysis, mathematical extension of results, and grant writing.
- *How Complex Systems Cope with Noise*, Army Research Office (F.R. Adler, PI, 2017). Literature synthesis on the generative role of stochasticity and decision-making under incomplete information.

Consultant (self-employed), Reno, NV and Austin, TX **2010–2016**
Interface design and documentation for a genomics pipeline platform (Lab7.io); security algorithm design for confidential genomic data (Genformatic LLC); an auditory interface for distinguishing nuclear tests from earthquakes (Comprehensive Nuclear-Test-Ban Treaty Organization).

Content Writer, Impact Mill (formerly eEcosphere), Austin, TX **2016**
Articles on complex environmental topics for a general audience.

Technical Writer, Spredfast, Austin, TX **2014**
Documentation and support strategy for two merging software companies.

Data Scientist, BetaZi Predictive Analytics, Truckee, CA **2011–2012**
Data pipeline, processing, and user-interface design for statistical analysis software.

Game Designer, International Game Technology (IGT), Reno, NV **2008–2010**
Designed and calculated the mathematics of casino games; led a team of 20+ artists and engineers.

SELECTED TRAINING

2026	Multi-scale Biological Adaptation to Environmental Variability (Workshop), NSF-Simons National Institute for Theory and Mathematics in Biology (NITMB), Chicago, IL (by invitation)
2024	Postdocs in Complexity Global Summit, Santa Fe Institute, Santa Fe, NM (by invitation)
2023	AMS Mathematics Research Community: Complex Social Systems, Buffalo, NY
2020	Adaptive Management Tutorial, National Institute for Mathematical and Biological Synthesis, University of Tennessee, Knoxville
2018	Data-Driven Modeling of Collective Behavior, Mathematical Biosciences Institute, Ohio State University (by invitation)
2017	Data-Driven Modeling of Collective Behavior and Emergent Phenomena in Biology, Statistical and Applied Mathematical Sciences Institute, Durham, NC (by invitation)